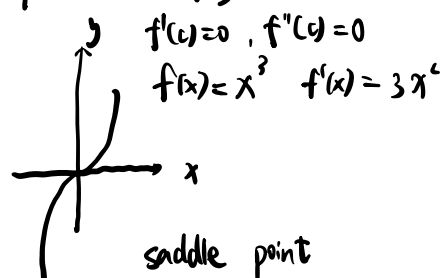
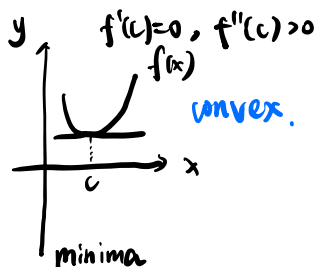
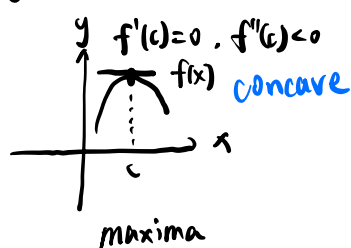


Lecture 8 Local Extrema

$f(x)$: If $f'(c) = 0$, $(c, f(c))$ is called critical point of $f(x)$



$f'(x)$ is decreasing & $f'(c)=0$

$\Rightarrow f'(x) > 0$ when $x < c \Rightarrow f(x)$ is increasing when $x < c$

$f'(x) < 0$ when $x > c \Rightarrow f(x)$ is decreasing when $x > c$

$z = f(x, y)$, D is the domain of f

Def. (Local Maxima/Minima) A function f has a local maxima at (a, b) if $\exists \delta > 0$ s.t. $\forall (x, y) \in D$ & $\sqrt{(x-a)^2 + (y-b)^2} < \delta$, we have $f(x, y) \leq f(a, b)$.

A function f has a local minima at (a, b) if $\exists \delta > 0$ s.t. $\forall (x, y) \in D$ & $\sqrt{(x-a)^2 + (y-b)^2} < \delta$, we have $f(x, y) \geq f(a, b)$.

$z = f(x, y)$. the equation of the tangent plane is

$$z = f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b)$$

If $f_x(a, b) = f_y(a, b) = 0$, then this plane is horizontal.

Def. (Critical Points) Given $f(x, y) \in C^1$. A point (a, b) is said to be a critical point if the tangent plane at (a, b) is horizontal.

$$(a, b) \text{ is a critical point} \iff \frac{\partial f}{\partial x}(a, b) = \frac{\partial f}{\partial y}(a, b) = 0 \iff \nabla f(a, b) = 0.$$

Critical points are candidates of local extrema

Example. Find all critical point(s) of the function $f(x,y) = \sin x \sin y$.

Sol. $\frac{\partial f}{\partial x} = \cos x \sin y = 0$ and $\frac{\partial f}{\partial y} = \sin x \cos y = 0$

$$(\cos x = 0 \text{ or } \sin y = 0) \text{ and } (\sin x = 0 \text{ or } \cos y = 0)$$

$$(x = \frac{\pi}{2} + k\pi \text{ or } y = m\pi) \text{ and } (x = n\pi \text{ or } y = \frac{\pi}{2} + p\pi)$$

$$\Rightarrow (x = \frac{\pi}{2} + k\pi \text{ and } y = \frac{\pi}{2} + p\pi) \text{ or } (x = n\pi \text{ and } y = m\pi)$$

$$\Rightarrow (\frac{\pi}{2} + k\pi, \frac{\pi}{2} + p\pi), (m\pi, n\pi) \text{ for any integers } k, p, m, n.$$

Second Derivative Test.

$$f(x,y) = x^2 + 4xy + y^2 = (x+y)^2 + 2xy$$

$$\nabla f(x,y) = \langle 2x+4y, 2y+4x \rangle$$

$$\nabla f(0,0) = \langle 0,0 \rangle \Rightarrow (0,0) \text{ is a critical point}$$

$$\frac{\partial^2 f}{\partial x^2} = 2 > 0, \frac{\partial^2 f}{\partial y^2} = 2 > 0 \quad \nRightarrow (0,0) \text{ is a local minima}$$

$$f(0,0) = 0$$

$$f(a, -a) = -2a^2 < 0, f(a, a) = 6a^2 > 0 \Rightarrow (0,0) \text{ is a saddle point.}$$

Thm (Second Derivative Test for Two-Variable Functions) Let $f(x,y) \in C^2$ and (x_0, y_0) is a critical point, i.e. $\nabla f(x_0, y_0) = \vec{0}$. Then the nature of this critical point (x_0, y_0) is determined by the following table

$(f_{xx}f_{yy} - f_{xy}^2) _{(x_0, y_0)}$	$f_{xx}(x_0, y_0)$	(x_0, y_0) is a
> 0	> 0	local minima
> 0	< 0	local maxima
< 0	anything	saddle

Any other cases are inconclusive

$$f(x,y) = x^2 + 4xy + y^2, \quad f_{xx} = f_{yy} = 2, \quad f_{xy} = 4$$

$$(f_{xx} f_{yy} - f_{xy}^2)|_{(0,0)} = 2 \times 2 - 4^2 < 0$$

$D(x,y) = f_{xx} f_{yy} - f_{xy}^2$ is called the discriminant of f
 is also the determinant of the Hessian matrix $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$.

Pf of Thm. $y = f(x), \quad f'(a) = 0$

Recall that the Taylor's expansion series

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 \quad \text{when } x \text{ is near } a$$

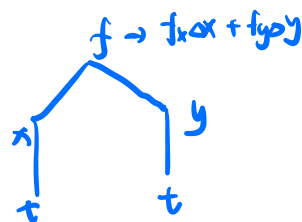
$$= f(a) + \frac{1}{2}f''(a)(x-a)^2$$

If $f''(a) > 0$, then $f(x) \geq f(a)$ local minima

If $f''(a) < 0$, then $f(x) \leq f(a)$, local maxima.

$$z = f(x,y), \quad \nabla f(x_0, y_0) = \vec{0}$$

$$\begin{cases} x(t) = x_0 + t\Delta x \\ y(t) = y_0 + t\Delta y \end{cases} \quad F(t) = f(x(t), y(t))$$



$$F'(t) = f_x x'(t) + f_y y'(t) = f_x \Delta x + f_y \Delta y$$

$$F''(t) = \frac{d}{dt} (f_x \Delta x + f_y \Delta y)$$

$$= \frac{\partial (f_x \Delta x + f_y \Delta y)}{\partial x} \cdot \Delta x + \frac{\partial (f_x \Delta x + f_y \Delta y)}{\partial y} \Delta y$$

$$= f_{xx} \Delta x^2 + f_{yx} \Delta x \Delta y + f_{xy} \Delta x \Delta y + f_{yy} \Delta y^2$$

$$= f_{xx} \Delta x^2 + 2f_{xy} \Delta x \Delta y + f_{yy} \Delta y^2$$

$$F(t) = F(0) + F'(0)t + \frac{1}{2}F''(0)t^2 + \text{higher order terms.}$$

$$\Rightarrow f(x(t), y(t)) \approx f(x_0, y_0) + f_x \Delta x + f_y \Delta y + \frac{1}{2} (f_{xx} \Delta x^2 + 2f_{xy} \Delta x \Delta y + f_{yy} \Delta y^2)$$

$$\Rightarrow f(x, y) \approx f(x_0, y_0) + \cancel{f_x(x_0, y_0)(x - x_0)} + \cancel{f_y(x_0, y_0)(y - y_0)}$$

$$+ \frac{1}{2} (f_{xx}(x_0, y_0)(x - x_0)^2 + 2f_{xy}(x_0, y_0)(x - x_0)(y - y_0) + f_{yy}(x_0, y_0)(y - y_0)^2)$$

$$A = f_{xx}(x_0, y_0), \quad B = f_{xy}(x_0, y_0), \quad C = f_{yy}(x_0, y_0)$$

$$X = x - x_0, \quad Y = y - y_0$$

$$AX^2 + 2BXY + CY^2 \quad \text{sign of this?}$$